NORTH SOUTH UNIVERSITY · DEPARTMENT OF ELECTRICAL & COMPUTER ENGINEERING (ECE) · CSE499 CAPSTONE

A Voice-Based AI Framework for Medical Pre-Screening of Bangla & Regional Bangladeshi Speech

with Intelligent Symptom Capture, Risk Assessment, Clinical Documentation & EHR Integration

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3 · From Spoken Symptoms to a Doctor-Ready Record — System Workflow

the patient's exact words are preserved end-to-end; the doctor makes every clinical decision

1. **Patient Speaks** — Symptoms by voice — Bangla / Banglish / dialect (typing fallback)
2. **Speech Understanding** — Transcribe → clean → normalise the spoken input
3. **Information Capture** — Extract symptoms; follow-up loop fills every gap
4. **Assessment & Triage** — Risk tier + rule-based red-flag check, explained
5. **Clinical Documentation** — Structured record → EHR; export FHIR · PDF · DOCX
6. **Doctor Review** — Verify, prescribe & decide → doctor-ready record

1 Introduction & Objective

- In busy Bangladeshi clinics, symptom history is taken verbally in **Bangla, Banglish or dialect**, under heavy time pressure — details are lost before consultation.
- Capturing this speech reliably is hard; a **raw transcript is neither complete nor structured**, and cannot serve as a clinical record.
- Voice-first interaction suits **elderly, low-literacy and non-technical** patients better than typed forms, and structured pre-consultation records save clinician time.

OBJECTIVE

- Capture symptoms **naturally by voice** (typing fallback) in Bangla / Banglish / dialect.
- Produce a **verified, structured** clinical summary and complete missing details through follow-up.
- Support clinical triage and **keep the doctor's review in the loop**.
- Stay dialect-aware and **preserve the patient's own words** for traceability.
- Generate a doctor-ready **EHR / FHIR** record — **assist, never diagnose**.

TARGET USERS

Patients · triage medics · doctors — one shared record across three portals.

EXPECTED IMPACT

Faster, more complete intake and fewer missed details, giving the doctor a structured record they can trust — without adding clinical risk or replacing judgment.

2 Problem, Motivation & Novelty

PROBLEM

Dialect variation, code-switching, medical terms & clinic noise make single-pass Bangla ASR unreliable — a transcript alone can't be trusted clinically.

MOTIVATION

Cutting documentation burden and missed information at intake directly improves consultations in high-volume, low-resource clinics.

NOVELTY

A **stateful, conversational** pre-screener that **verifies & fills gaps** instead of trusting one transcript — with rule-based red-flag safety, **explainable** risk, and a dialect-aware, EHR/FHIR-integrated **three-portal, human-in-the-loop** workflow.

KEY CONTRIBUTION

An end-to-end, three-portal working prototype (voice → structured record → EHR/FHIR) **plus** a 16-model Bangla dialect ASR benchmark whose findings shaped the conversational, verify-and-complete design.

WHY IT MATTERS

In high-volume, low-resource clinics, a few minutes of clean, structured history per patient compounds into meaningful time saved and safer consultations at scale.

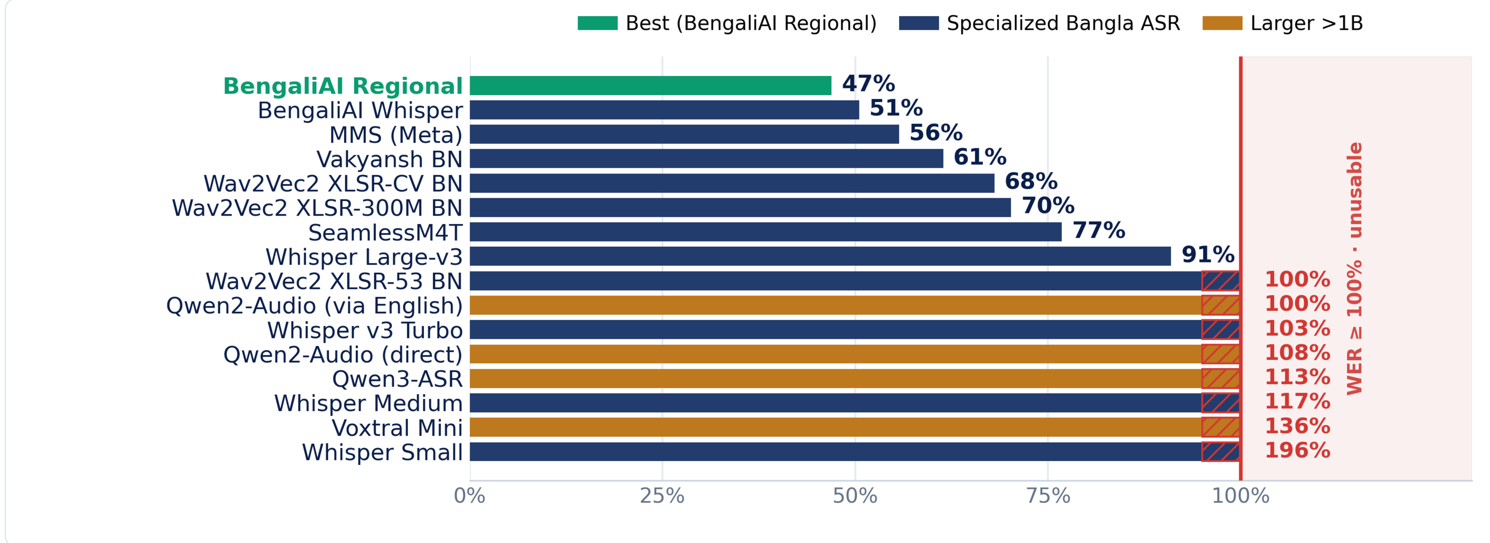
4 Methodology — how the system works

1. **Input — Patient Voice**
   Patient speaks symptoms in Bangla / Banglish / dialect (typing fallback); the browser captures the audio.
2. **Speech Capture & Recognition**
   Web Speech API (bn-BD) transcribes speech to text; the **raw transcript is stored unchanged**.
3. **Confirmation & Correction**
   Text is cleaned / normalised in a separate field and shown back to confirm — raw words are never overwritten.
4. **Structured Information Capture**
   An LLM extracts symptoms, duration, severity & history into a **10-field summary**; missing fields trigger spoken follow-up questions, looping until complete.
5. **Clinical Assessment & Triage**
   Risk graded Low→Critical with a rule-based **red-flag check** + a plain-language reason; the medic verifies fields, records vitals & triages.
6. **Doctor Review**
   The doctor sees the safety panel, history & record and accepts, overrides or prescribes — the clinical decision stays human.
7. **Clinical Record / EHR**
   A structured, no-diagnosis report is stored in the EHR & exported as **FHIR / PDF / DOCX**.

DATA & EVALUATION

~**4.7 h** multi-dialect audio (16 kHz mono) collected & preprocessed; speech models scored by **WER / CER / BLEU** per dialect. One small service per module; every model call logs its provider & latency, and each risk output carries a stored reason.

5 Speech-Recognition Investigation

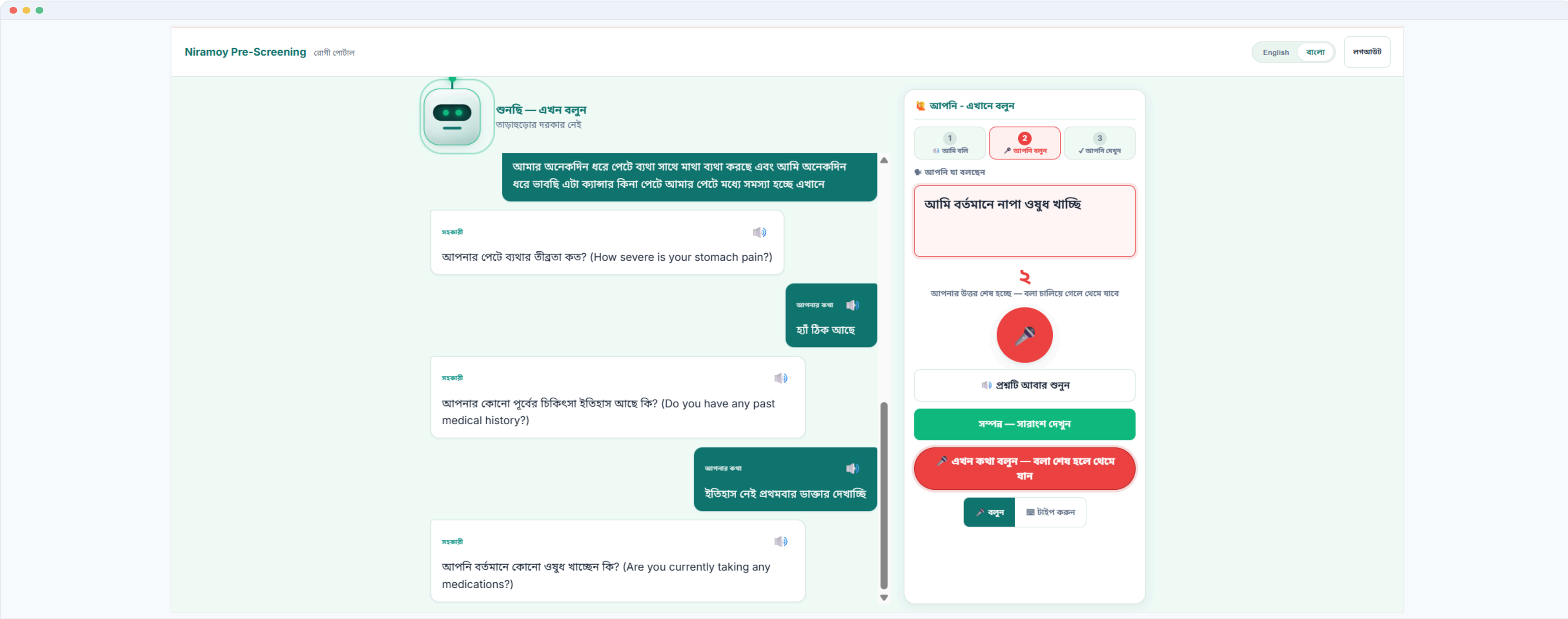


16 ASR systems benchmarked on multi-dialect medical speech. **WER** counts substitutions, deletions **and insertions** ÷ words — so hallucinated extra words push WER **past 100%** (more errors than words = unusable). Best model still errs ~47%; larger models did **not** help.

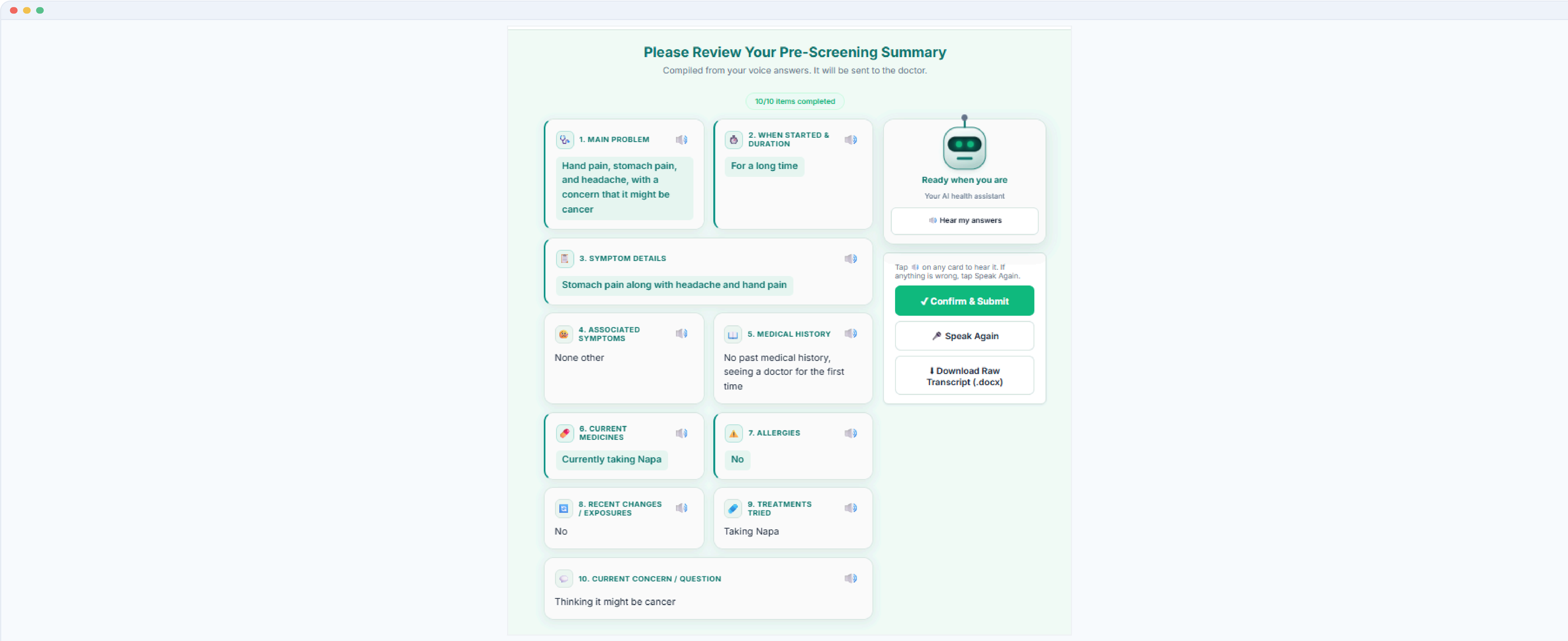
Honesty note: this is an **offline research benchmark**; the live prototype uses the browser Web Speech API, **not** these models.

16	47%	8
models tested	best WER	failed (≥100%)

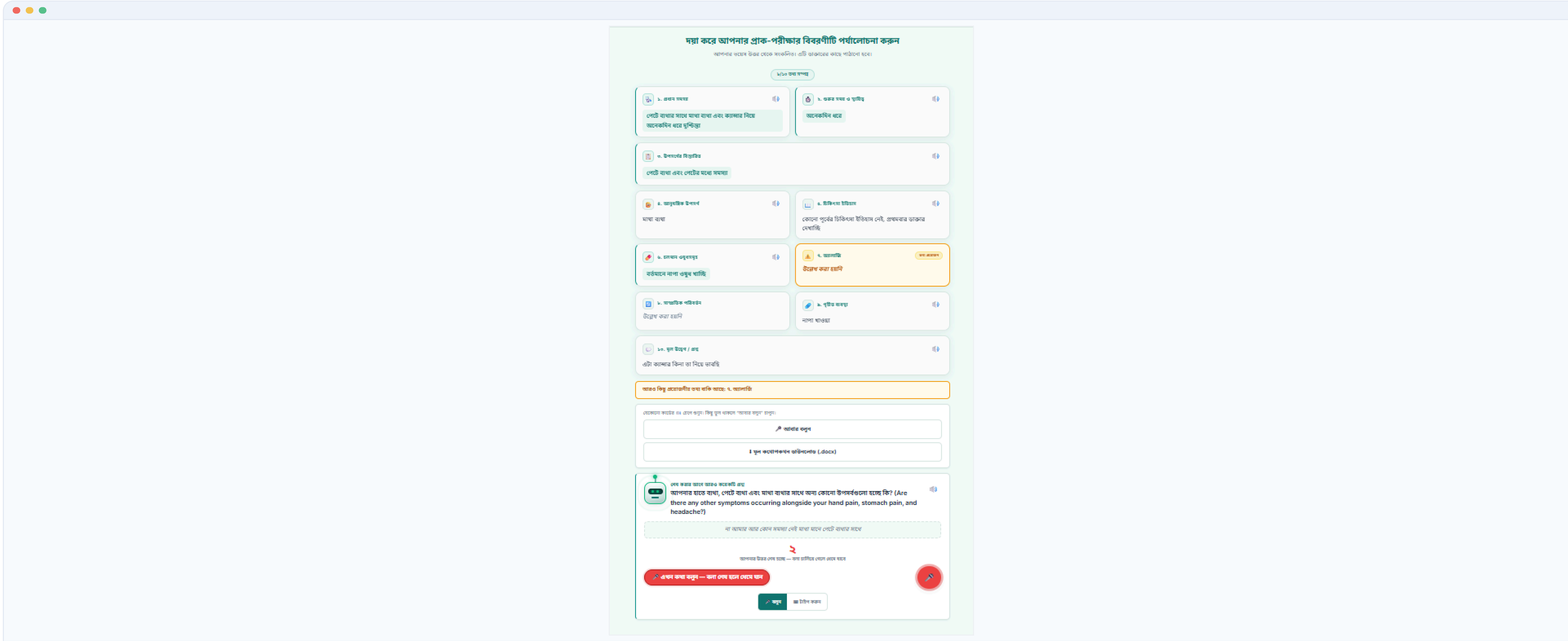
6 System Features — the Three Portals & the Clinical Record



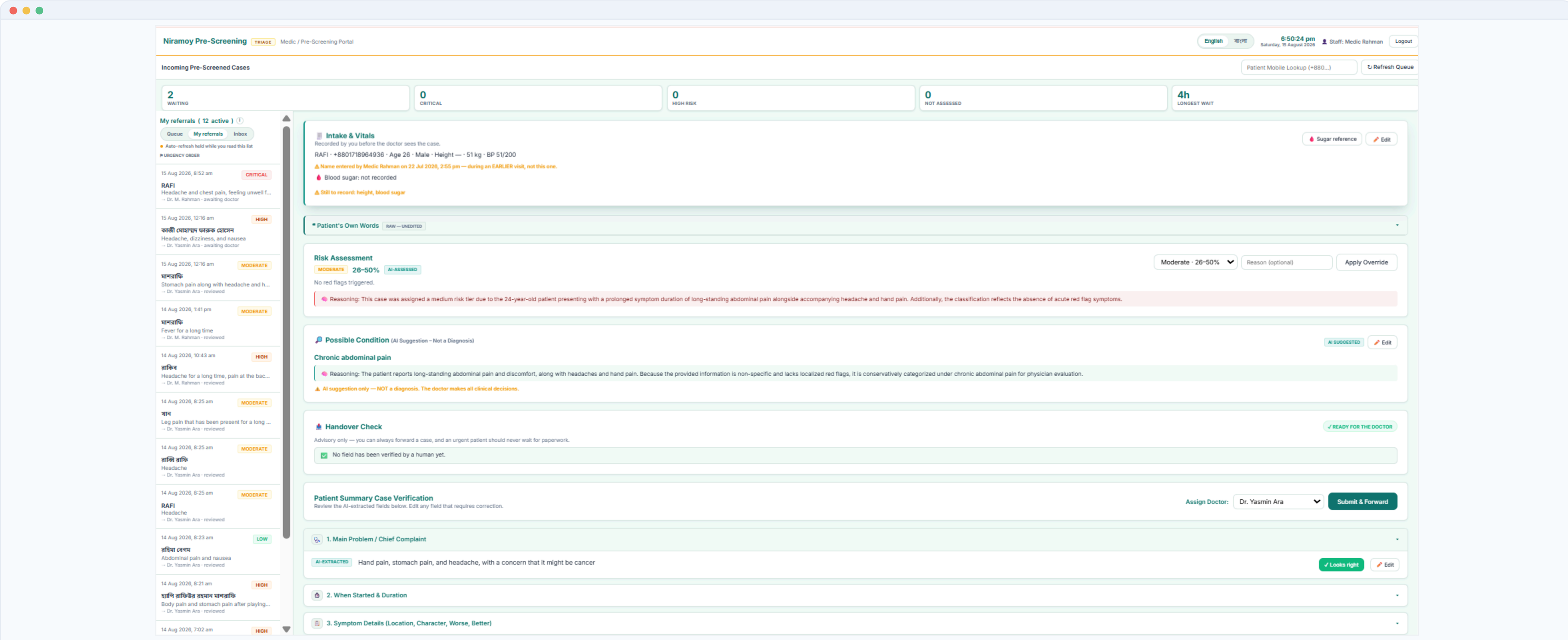
Patient Portal — Voice-First Capture
The patient answers spoken follow-up questions in Bangla by voice (or typing); their exact words are kept and each field is confirmed — natural, dialect-friendly intake for non-technical patients.



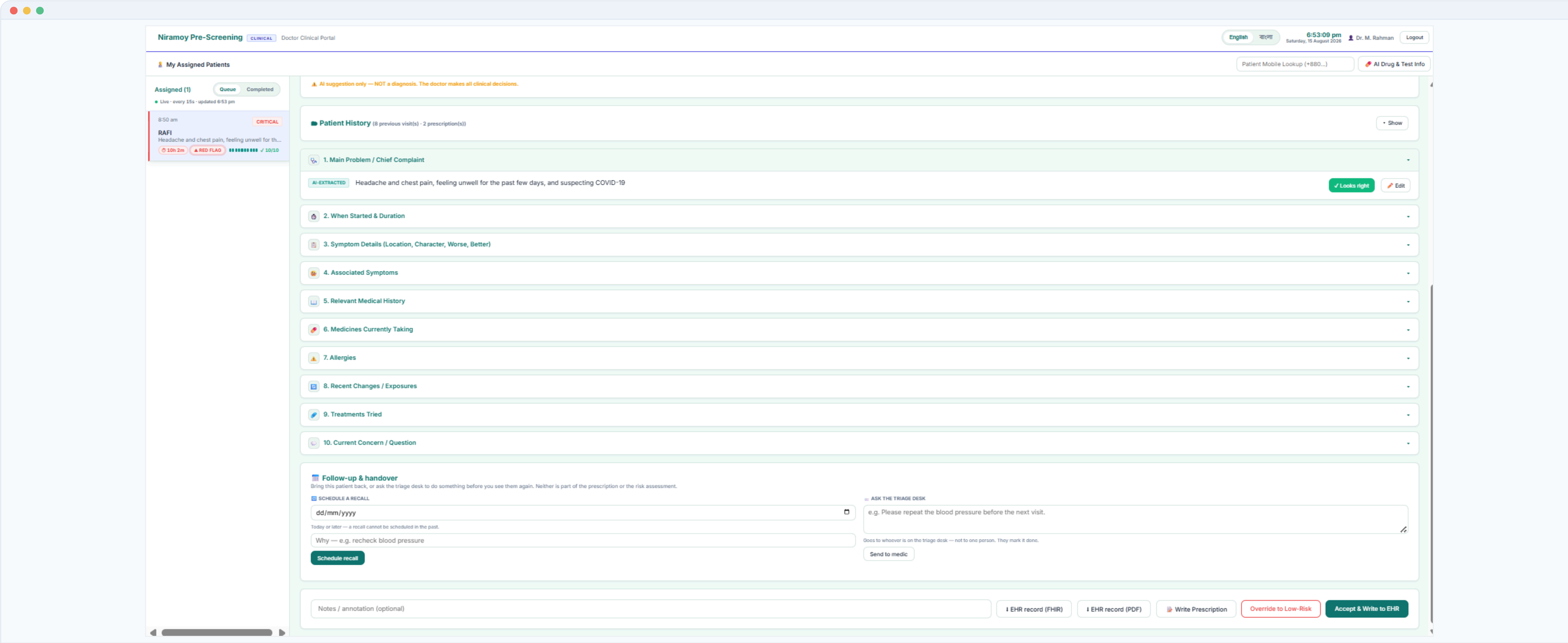
Patient Portal — Review & Confirm
The 10 captured fields are shown back before submission. The patient verifies and corrects the summary, improving data quality and giving informed control over their own record.



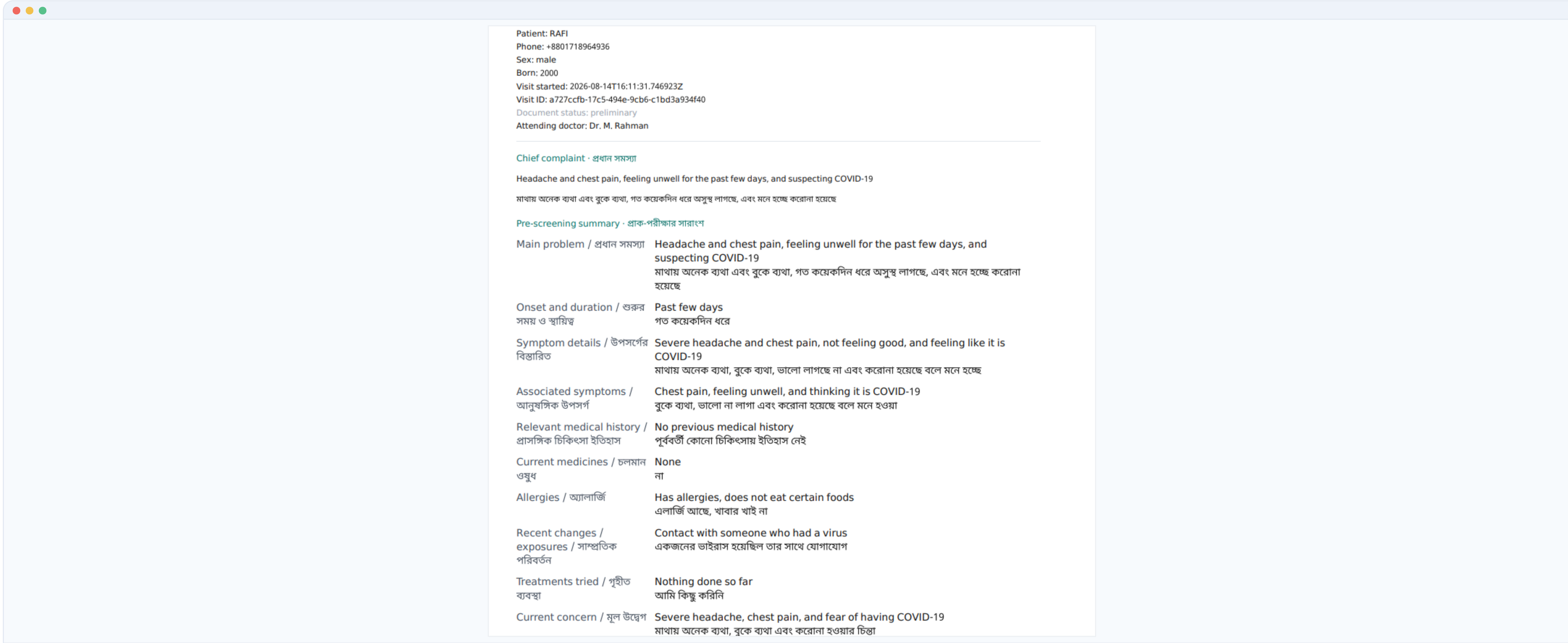
Patient Portal — Bangla Review
The same confirmation screen, fully bilingual, keeps the system usable for non-English-speaking and low-literacy patients — central to the Bangladeshi use case.



Medic Portal — Triage & Verification
Cases are ordered by risk with the AI's reasoning shown; the medic records vitals, verifies the extracted fields and forwards to a doctor — ensuring urgent patients are seen first.



Doctor Portal — Review & Decision
Safety panel (risk tier, red flags, explanation), patient history and EHR / prescription actions. The doctor accepts, overrides or prescribes — the final clinical decision stays human.



Clinical Record — EHR / FHIR Output
A structured, bilingual pre-screening document is stored in the EHR and exported as HL7 FHIR R4, PDF & DOCX — a doctor-ready record to attach to the consultation.

7 Technology Stack

- **BACKEND**: Python · FastAPI · REST + WebSocket
- **SPEECH / VOICE**: Web Speech API (bn-BD) · edge-tts
- **AI / LLM**: Gemini · Groq · OpenRouter · + fallback
- **CLINICAL DATA & EXPORT**: SQLite + Alembic · HL7 FHIR R4 · fpdf2 · python-docx
- **FRONTEND**: HTML · CSS · JS · 3 portals
- **TESTING / INFRA**: pytest · Google Colab

8 Limitations & Future Work

LIMITATIONS

- Live STT relies on the cloud browser API; dialect recognition stays error-prone.
- Login / OTP & encryption are prototype-stage — not production security.
- Extraction accuracy not yet formally measured; depends on free-tier LLMs.
- Some regional dialects had limited evaluation data.

FUTURE WORK

- On-device **quantised** STT & summary model — offline & private.
- Fully hands-free voice follow-up; LoRA dialect ASR fine-tuning.
- Formal accuracy evaluation (extraction precision / recall).
- Production authentication, encryption & real SMS OTP.

9 Conclusion

A complete **voice-first, safety-first** prototype turns spoken Bangla / dialect symptoms into a structured, doctor-ready **EHR / FHIR** record across three portals, backed by a 16-model ASR study. It **reduces intake burden and organises clinical information while keeping clinical judgment with humans** — it assists the workflow and **never diagnoses**.